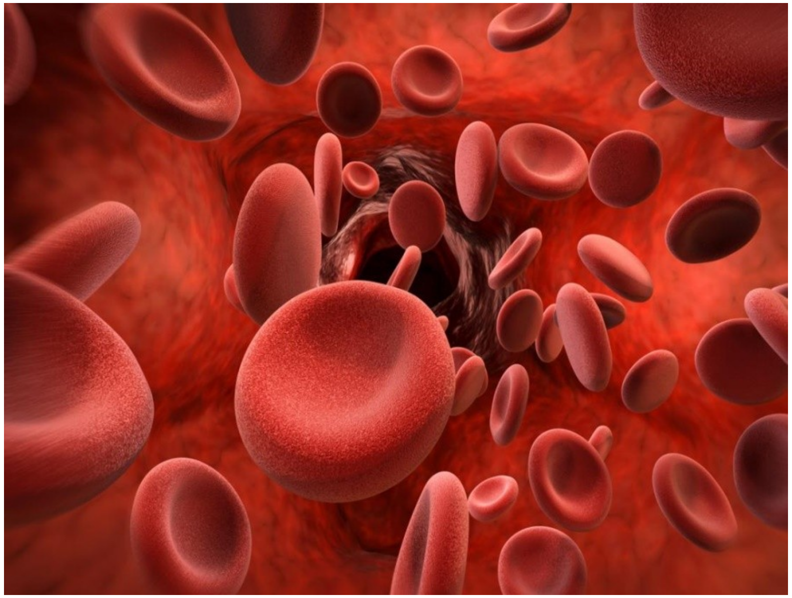
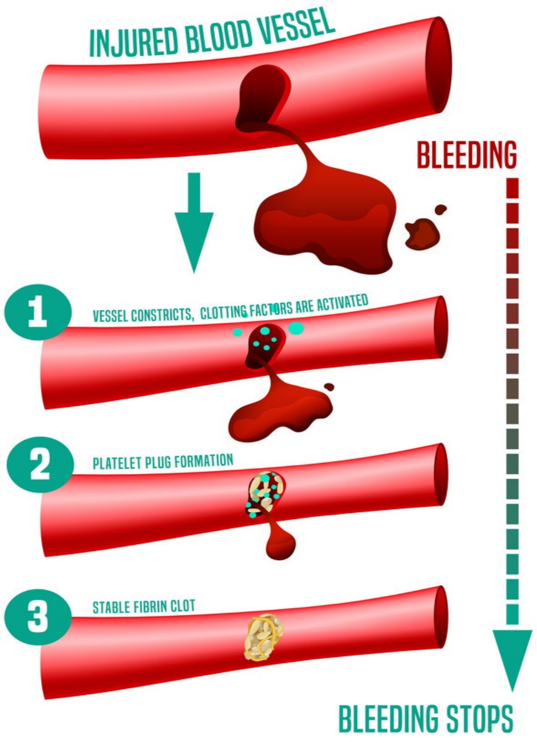


Bleeding disorders

Dr. Hanaa Elsayed

Learning Outcomes

- By the end of this presentation you should be able to:
  - Define Hemostasis.
  - Discuss disorders of primary hemostasis
  - Describe the clinical picture of disorders of primary hemostasis
  - Explain investigations available to reach diagnosis
  - Outline management of ITP
  - Discuss disorders of secondary hemostasis

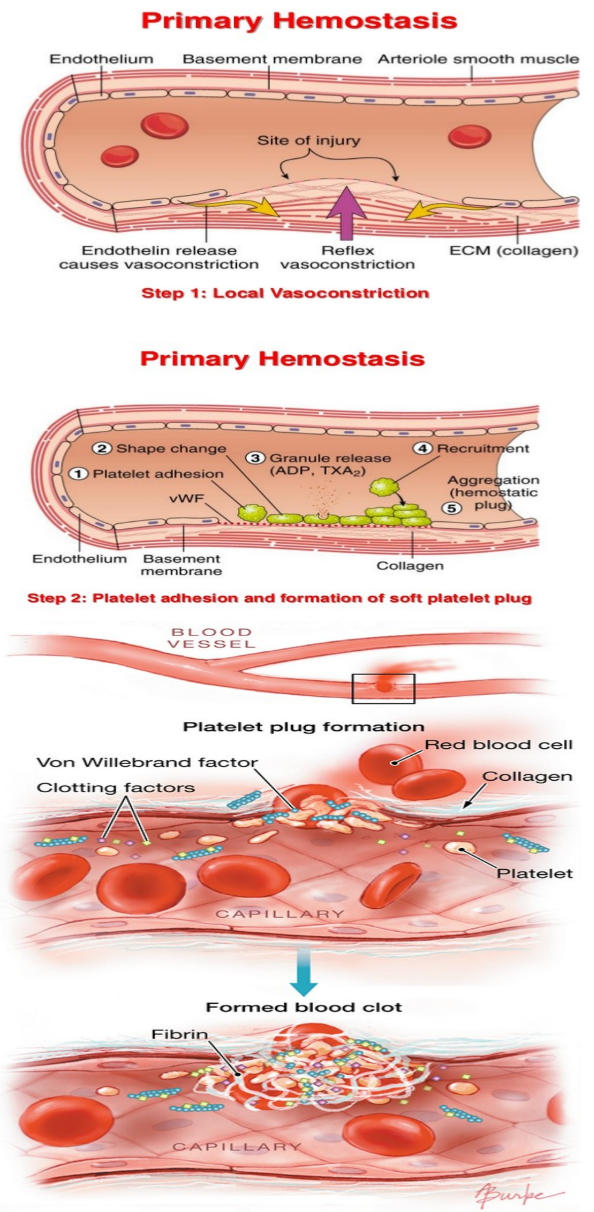


Introduction

- **Definition Of Hemostasis:**
  - **It is a process to prevent and stop bleeding.**
    - **Normal hemostasis depends on interaction between blood vessel wall, platelets and coagulation cascade.**
    - **The endothelial cells of intact vessels prevent blood clotting with a heparin-like molecule and thrombomodulin and prevent platelet aggregation with nitric oxide and prostacyclin.**
    - **When endothelial injury occurs, the endothelial cells stop secretion of coagulation and aggregation inhibitors and secrete von Willebrand factor which initiate the maintenance of hemostasis after injury.**
- **Three reactions- primary, secondary and tertiary hemostasis-act simultaneously.**

Primary hemostasis

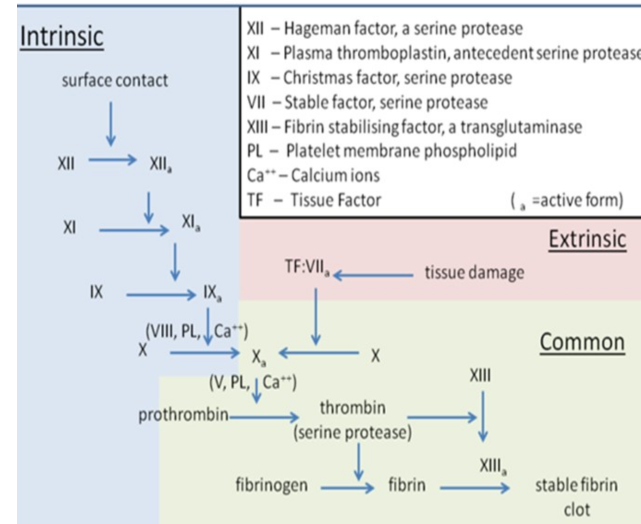
- After initial injury there is a brief period of arteriolar vasoconstriction mediated by:
  - Reflex neurogenic mechanisms and augmented by local secretion of factors such as endothelin (a potent endothelium derived vasoconstrictor).
- A primary platelet plug forms by platelet adhesion mediated by von Willebrand factor (vWF).
- Activated platelets release contents of their granules (contain a variety of substances) that stimulate further platelet activation and enhance the hemostatic process.



Secondary hemostasis

- It is the formation of fibrin through coagulation cascade.
- There is 3 pathways; intrinsic, extrinsic and common.
- Extrinsic pathway:
  - It is the main pathway.
  - It is initiated by the generation/exposure of tissue factor (factor III or thromboplastin) at the site of injury and the tissue factor–factor VII complex, the latter activating factor X.

The three pathways that makeup the classical blood coagulation pathway



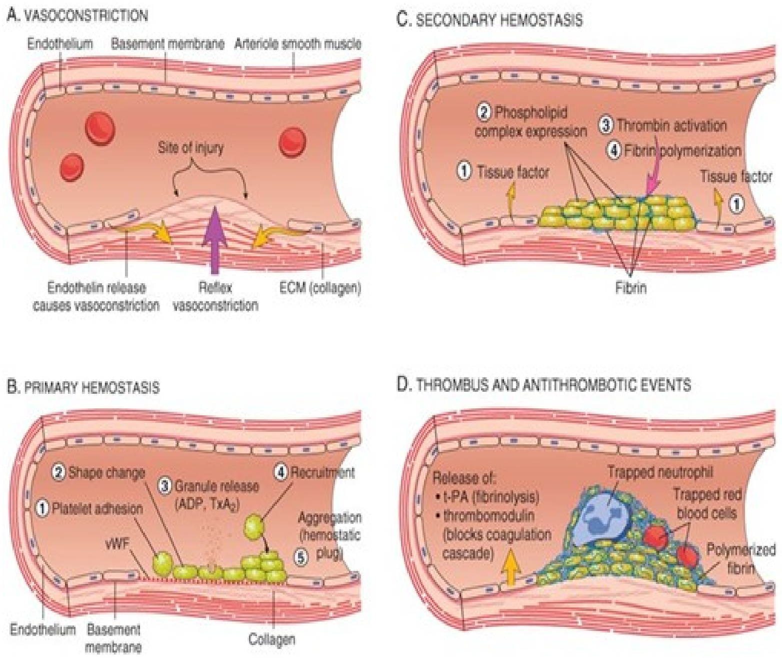
Clotting Factors

Factors	Names
I	Fibrinogen
II	Prothrombin
III	Thromboplastin (tissue factor)
IV	Calcium
V	Labile factor
VII	Stable factor
VIII	Antihemophilic factor A
IX	Antihemophilic factor B
X	Stuart-Prower factor
XI	Plasma thromboplastin antecedent (PTA)
XII	Hageman factor
XIII	Fibrin stablizing factors

- **Intrinsic pathway:**
  - It begins with formation of complex on collagen by high-molecular-weight kininogen (HMWK), prekallikrein, and factor XII (Hageman factor).
  - This initiates a cascade in which factor XII is activated, which then activates factor XI, which activated factor IX, which along with factor VIII activates factor X.
- **Common Pathway:**
  - When factor X is activated by either the intrinsic or extrinsic pathways.
  - Activated factor X activates prothrombin (factor II) and converts it into thrombin.
  - Thrombin then cleaves fibrinogen into fibrin, which forms the mesh that binds to and strengthens the platelet plug.

Tertiary hemostasis (Fibrinolysis)

- It is the breakdown of a fibrin clot.
- This is critical to the process of wound healing, in order to restore vessel patency and for tissue remodeling.
- Tissue-type plasminogen activators (t-Pas) is released from the endothelial cells and converts plasminogen to plasmin, which degrades fibrinogen and fibrin into fibrin degradation products (FDPs).



Disorders of Primary Hemostasis

- **Definition:**
  - **Inability to form an adequate platelet plug due to:**
    - **Disorders of blood vessels.**
    - **Disorders of platelets.**
      - » **Abnormal function.**
      - » **Abnormal numbers (thrombocytopenia).**
    - **Disorders of vWF.**
- **Results in spontaneous bleedings affecting skin (petechial, purpuric rash, ecchymosis) also affect mucus membranes (gingiva, epistaxis, and orificial bleeds).**
- **Prolonged bleeding time after trauma (common).**
- **Intra-cerebral and intra-abdominal bleedings can also occur.**

Vascular Bleeding Disorders

- Vascular purpura rarely causes serious bleeding.
- Clinical pictures: bleeding from mucous membranes or skin starts immediately after trauma but stops within 24–48 hours.

Figure 2 Telangiectases on the tongue of an 85-year-old man with hereditary hemorrhagic telangiectasia who suffered from occasional episodes of epistaxis.

- Platelets are normal count and function.
- Causes:
  - Hereditary hemorrhagic telangiectasia.
  - Senile purpura.
  - Purpura associated with infections e.g. infective endocarditis and vasculitis.

Hereditary Hemorrhagic Telangiectasia (HHT)

- **Autosomal dominant condition.**
- **There is vascular dysplasia leading to telangiectasia and arteriovenous malformations (skin or mucosa of oral, nasal, conjunctiva, respiratory, gastrointestinal or urogenital, or brain and liver leads to bleeding).**
- **Clinical diagnostic criteria at least 3 of 4 of the following criteria:**
  - **Spontaneous and recurrent epistaxis.**
  - **Multiple telangiectasia at lips, oral cavity, nose, fingers.**
  - **Visceral arteriovenous malformations.**
  - **Positive family history.**

- **Investigations:**
  - **Capillary microscopy, CT, MRI and angiography.**
- **Treatment:**
  - **Cryosurgery or laser ablation, electrocoagulation, or antifibrinolytics.**

Dental aspects of Telangiectasia

- **Telangiectases may be seen in any part of the mouth and may be conspicuous (obvious) on the lips.**
- **Bleeding from oral surgery.**
- **Antibiotic prophylaxis should be considered before dental care to reduce the occurrence of cerebral abscesses for patients with HHT and concomitant pulmonary arteriovenous malformation.**
- **Regional LA should be avoided because of the risk of deep bleeding.**
- **Conscious sedation can be given if required.**
- **GA, nasal intubation is best avoided and close post-operative observation is advisable.**

Platelet Abnormalities

A- Quantitative platelet abnormalities (Thrombocytopenia):

- **Decreased production/maturation:**
  - **Idiopathic (autoimmune thrombocytopenia).**
  - **Bone marrow infiltration (leukemia, or carcinoma), Viruses such as cytomegalovirus and HIV and Drugs such as cytotoxic.**
- **Increased platelet destruction:**
  - **Idiopathic Thrombocytopenic purpura (ITP).**
  - **Acquired:**
    - **Hemolytic uremic syndrome (HUS) and thrombotic thrombocytopenic purpura (TTP).**
    - **Hypersplenism (splenomegaly) caused by:**
      - » **Portal hypertension e.g. cirrhosis.**
      - » **Hematological disorders e.g. myeloproliferative disorders.**
      - » **Infections, e.g. malaria.**
      - » **Inflammatory disorders, e.g. systemic lupus erythematosus (SLE).**

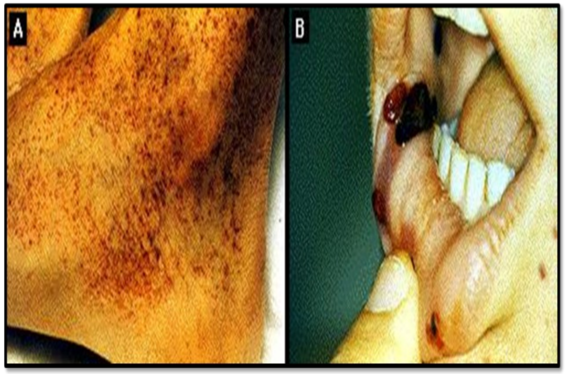
B- Qualitative Platelets Abnormalities

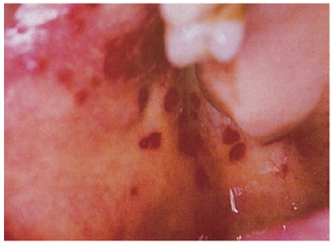
- **Congenital:**
  - **Glanzman's thrombasthenia (may be acquired in association with pregnancy, autoimmune).**
    - **Treatment of oral surgical bleeding involves platelet transfusion, antifibrinolytics, recombinant factor (rF) VIIa, and local hemostatic agents, alone or in combination..**
  - **von Willebrand disease.**
- **Acquired: as seen with drugs such as aspirin, clopidogrel, NSAIDS, and many antibiotics.**

Platelet defects	The purpuras[a]	Coagulation defects
Gender affected	Females mainly	Males
Family history	Rarely	Usually positive
Nature of bleeding after trauma	Immediate	Delayed
Effect when locally applied pressure removed	May stop bleeding	Bleeding recurs
Spontaneous bleeding into skin or mucosa, or from mucosa or gingivae	Common	Uncommon
Bleeding from minor superficial injuries (e.g. needle prick)	Common	Uncommon
Deep haemorrhages or haemarthroses	Rare	Common
Bleeding time	Prolonged	Normal
Tourniquet (Hess) test	Positive	Negative
Platelet count	Often low	Normal
Clotting function	Normal	Abnormal

Idiopathic or primary or immune thrombocytopenic purpura (ITP)

- It is due to autoantibody- mediated destruction of platelets.
- Viral infections as HIV, HCV.
- Clinical features:
  - Easy bruising, petechial, purpura, epistaxis, menorrhagia and postoperative hemorrhage are common.
  - Major hemorrhage is rare and is seen only in patients with severe thrombocytopenia.
  - Physical examination:
    - Normal except for evidence of bleeding.
    - Splenomegaly is rare.





Management

- **History and examination.**
- **Complete Blood count (CBC) and film show the number and morphology of platelets.**
- **Bleeding Time (normal 3-10 min.) is prolonged.**
- **HIV, HCV (if risk factors are present)**
- **Treatment: Treat cause.**
  - **First line of treatment: Corticosteroids.**
  - **Second line:**
    - **Splenectomy: vaccinations prior (Pneumococcus, Meningococcus, H. influenza B).**
    - **Immunosuppressant (azathioprine, cyclophosphamide).**
  - **IVIg 1 g/kg/d x 2 doses, or 2 g/kg over 5 d**
  - **Platelet transfusion: for life-threatening bleeding.**
  - **Antifibrinolytic: tranexamic acid if refractory bleeding.**

Dental Aspects of Platelet Disorders

- The main danger is hemorrhage but it is rarely severe as in the clotting disorders.
- According to severity of platelet disorders, type of dental procedures:

Platelet count (× 109/L)	Severity of thrombocytopenia	Manifestations	Management in relation to type of surgery	
			Dento-alveolar	Maxillofacial
100–150	Mild	Mild purpura. Slight prolonged postoperative bleeding	• No platelet transfusion. • Local hemostatic measures. • Observe.	AS dento-alveolar.
50–100	Moderate	Purpura, postoperative bleeding.	• Platelets may be needed. • Local hemostatic measures. • Postoperative tranexamic acid mouth wash for 3 days.	• AS dento-alveolar.
30–50	Severe	Purpura, postoperative bleeding, even from venipuncture.	• Platelets needed. + • As above.	As above + Avoid surgery where possible.
<30	Life-threatening	Purpura, spontaneous bleeding.	• As above + • Avoid surgery where possible.	AS above.

- **Regional LA block injections:** give if platelet levels are above 30× 109/L.
- **Local hemostasis** after **dento-alveolar surgery** is usually adequate if platelet levels are above 50 × 109/L.
- **Major surgery** requires platelet levels above 75 × 109/L.
- **Conscious sedation** given but care if given by intravenous (IV) route.
- **GA** should be given in hospital, with expert intubation to avoid bleeding into airway.
- **Platelets replacement** by transfusions, but sequestration of platelets is very rapid.

- **When platelets given prophylactically: platelets should be given– half just before surgery to control capillary bleeding and half at the end of the operation to facilitate placement of adequate sutures.**
- **Platelet infusions carry risk of iso-immunization and blood-borne infection.**
- **The platelet transfusions can be reduced by local hemostatic measures and use of desmopressin or tranexamic acid or topical administration of platelet concentrates.**

- **Absorbable hemostatic agents** such as **oxidized regenerated cellulose** (Surgicel), synthetic collagen (Instat) or **microcrystalline collagen** (Avitene) may be **put in the socket** to assist clotting.
- **Drugs:** antiplatelet (aspirin), other NSAIDs and antibiotics (amoxicillin, cephalosporin, azithromycin) **should be avoided**.

Management of post-operative bleeding

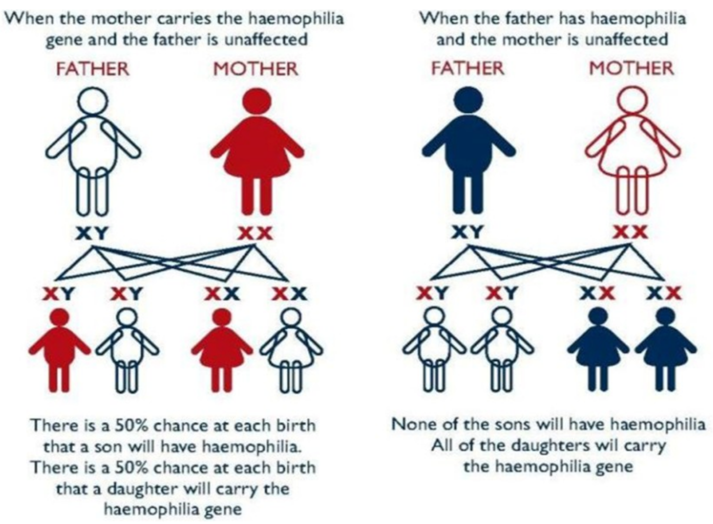
- Prolonged post-operative bleeding is defined if any of the following:
  - Bleeding continues for 12 hours.
  - Bleeding causes the patient to return for attention.
  - Bleeding causes a large hematoma/ecchymosis.

- Treatment:
  – The patient needs a transfusion.
  – Discovered site of by cleaning out the mouth with swabs.
  – Pressing firmly with a gauze pad for 10–15 min., usually stop bleeding even in some bleeding tendencies but often only temporarily.
  – Local hemostatic measures or Absorbable hemostatic agents.
  – Suture the socket under LA.

- If the bleeding persists, consider a systemic cause such as:
  – Acquired deficiencies of hemostasis, such as anticoagulants or thrombocytopenia, or
  – Hereditary deficiencies of clotting factors (e.g. von Willebrand syndrome and hemophilia).

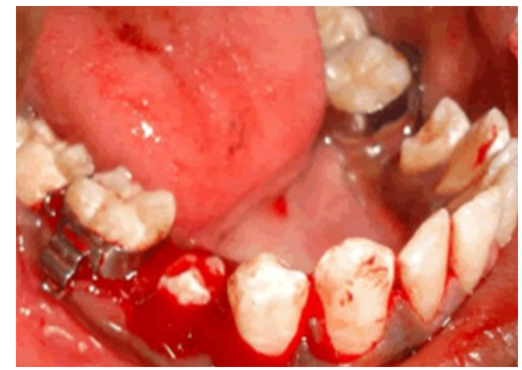
Blood coagulation disorders

- **Causes of coagulation disorders:**
  - **Drugs: anticoagulants (e.g. warfarin) and thrombolytic.**
  - **Liver failure and renal failure.**
  - **Genetic defects of clotting factors (vWD, hemophilia).**
- **Hemophilia A (X-linked, with ↓ factor VIII):**
  - **It is a rare genetic disorder (X-linked) (transmit to half of her sons and carrier state to half of her daughters).**
  - **It is occur due to Factor VIII deficiency, leading to excessive bleeding.**
  - **Affects males.**
  - **More common than Hemophilia B.**



Clinical pictures

- **Excessive bleeding from multiple site, particularly after minor trauma or (surgery) and sometimes spontaneously (severe condition).**
- **Hemorrhage stop immediately after the injury (due to normal vascular and platelet hemostatic responses).**
- **Hemorrhage is dangerous either due to acute blood loss or due to bleeding into tissues, particularly the brain, larynx, pharynx and muscles (hematoma).**



Oral Manifestations of hemophilia

- **Bleeding of the joints (hemarthrosis) is seen less frequently in temporomandibular joint and has a high risk of developing joint disease (arthropathy).**
- **Oral ulceration and ecchymosis of lip and tongue are commonly seen in toddlers.**
- **Post-traumatic gingival hemorrhage, gingival diseases.**
- **Bleeding site are labial frenum 60%, tongue 23%, buccal mucosa 17%, gingiva and palate 3%.**

Management

- **Diagnosis:**
  - **Prolonged activated partial thromboplastin time APTT (normal is 30-50 s).**
  - **Normal PT (16-18 s), INR is 1 (prolonged with abnormalities of extrinsic pathway, liver disease, or if the patient is on warfarin).**
  - **Low factor VIII.**
- **Treatment:**
  - **Desmopressin (synthetic vasopressin) are used for mild cases.**
  - **Factor VIII concentrate in severe cases or in the presence of bleeding.**
  - **In the absence of concentrates, cryoprecipitate can be used (risk of viral infections are very high).**
    - **World federation of hemophilia recommends cryoprecipitate or fresh frozen plasma to minimize the risk of bleeding.**

Other Congenital coagulation deficiency

Hemophilia B (Christmas disease):

- **As Hemophilia A but less common.**
- **Cause: factor IX deficiency.**
- **Management as Hemophilia A, but using factor IX and not using desmopressin.**

- **von Willebrand disease (vWD):**
  - **Most common type of hereditary coagulation disorders.**
  - **Autosomal dominant affecting males or females.**
  - **Cause: quantitative or qualitative deficiency of vW factor.**
  - **It is a protein needed for platelet adhesion and aggregation at injured site and is also carrier of factor VIII.**

- **Clinical pictures vWD:**
  - **Similar to platelet dysfunction; however, it can resemble to hemophilia.**
- **Pattern of clinical features in vWD is distinct from that of hemophilia.**
  - **Spontaneous and excessive bleeding**
    - **Bruising, epistaxis, gingival and prolonged bleeding during surgical or dental procedures are the most common features but menorrhagia or GIT bleeding also frequently.**
  - **Prolonged postextraction bleeding.**
  - **Post-traumatic gingival hemorrhage**
  - **Increased dentinal caries and periodontal diseases**
  - **Hemarthroses are possible but rare.**

- **Diagnosis:**
  - **PTT is prolonged because of the lack of availability of factor VIII.**
  - **Prolonged bleeding time resulting from platelet dysfunction.**
- **Treatment as hemophilia:**
- **(Depend on severity of disease, dental procedures, response of treatment):**
  - **Desmopressin and vWF replacement are usually the lines of therapy**

	Haemophilia A	Von Willebrand disease
Inheritance dominant	Sex-linked recessive	Autosomal
Haemarthroses/deep haematomas	Common	Rare
Epistaxes	Uncommon	Common
Gastrointestinal bleeding	Uncommon	Common
Haematuria	Common	Uncommon
Menorrhagia	None (males)	Common
Post-extraction bleeding	Starts 1–24 h after trauma, lasts 3–40 days and is not controlled by pressure	Starts immediately, lasts 24–48 h and is often controlled by pressure
Bleeding time	Normal	Prolonged
Factor VIII coagulant activity	Low	Low

Dental Aspects

- Hemophiliac patient should be minimize the need for dental intervention, which can cause severe, or occasionally fatal complications.
- Dental management should be focused on prevention, dietary counseling and periodic recalls in order to avoid invasive dental treatments.
- Preventive dentistry, should be started as early as possible in the young child, when the teeth begin to erupt.
  - Routine dental care starts with the beginning of tooth eruptions.
- Dental assessment is needed at an early stage to predict any future problems such as overcrowding.

- **Increases dental caries in hemophilic patient due to poor oral hygiene (difficulty in brushing).**
  - **Good oral hygiene by:**
    - **Regular oral hygiene instructions.**
    - **Brushing twice daily with a medium-texture brush is essential.**
    - **Use of fluoride tablets gels and mouth rinses, dietary sugar restriction and regular dental check-ups.**
- **All measure taken to reduce the risk of infection, soft tissue trauma and postoperative complications.**
- **HIV and hepatitis are closely associated with hemophilia and can have additional oral problems and opportunistic infections.**

- **Consult the hematologist before initiating the treatment.**
- **The management of hemophilia A depends on severity and the invasiveness of the dental procedure.**
  - **Mild to moderate hemophilia:**
    - **Nonsurgical dental treatment can be carried out under antifibrinolytic cover but with mandatory consultation with hematologist.**
    - **For mild hemophilia, scaling and minor surgical procedure can be carried out with desmopressin cover; however, the drug is ineffective in hemophilia B.**
  - **Severe hemophilia:**
    - **Factor VIII replacement therapy before procedure such as administration of local anesthesia, scaling or minor surgical procedure.**

- **Local anesthesia (80% chance of hematomas) regional blocks, lingual infiltrations or injections into the floor of the mouth should be avoided in the absence of factor VIII replacement. Intraligamental or intraosseous is considered to be a potential alternatives to blocks.**
- **Intramuscular injections of drugs should be avoided as they can cause large hematomas.**
- **Conscious sedation using inhalation sedation or midazolam is usually safe in selected cases.**
- **Give prophylactic antibiotic cover before any invasive procedure.**

- **Endodontic treatment preferred over surgical extractions.**
- **There is no contraindication to orthodontic treatment in hemophilia. Care must be taken with any sharp edges to orthodontic appliances or wires.**
- **Use of fibrin glue and tranexamic acid pre- and postsurgical procedure to control bleeding.**
- **Tranexamic acid when used topically as a rinse reduces bleeding following dental treatment.**

- **Following dental extraction:**
  - **Use Local hemostatic measures and other local measure such as fibrin glue and oxidized cellulose.**
  - **Oral antibiotics only prescribed if clinically necessary.**
  - **Avoid hot food, drinks and smoking.**
- **Extractions and dento-alveolar surgery should be carefully planned.**
  - **Factor VIII level of 50–75% .**
  - **Factor VIII and/or antifibrinolytic as Tranexamic acid (24 h preoperatively until 7th postoperative day)**
  - **Local hemostatic measures postoperative.**
- **When maxillofacial surgery is scheduled:**
  - **Factor VIII should be at a level of 75–100%.**
  - **Factor VIII (preoperatively).**

- **Post-operatively (5-10 days):**
  - **Local hemostatic measures.**
  - **Patients should be instructed to take a soft diet.**
  - **Report any swelling, swallowing difficulty or hoarseness to doctor immediately as may indicate hematoma formation.**
  - **Oral antibiotics only prescribed if clinically necessary.**
  - **If there is bleeding during this period repeat dose of factor VIII.**
- **For pain:** **paracetamol (acetaminophen) or a combination of codeine/paracetamol may be suitable. While NSAIDs and aspirin should be avoided.**

Acquired Coagulation Disorders

- **Acquired hemorrhagic disorders are far more prevalent than congenital diseases but, except in anticoagulant therapy or liver disease, are usually less severe.**
- **Causes include:**
    - **Anticoagulant.**
    - **Liver disease:**
        - » **Liver failure leads to impaired synthesis of vitamin K-dependent clotting factors (II, VII, IX, X and fibrinogen) – leading to prolonged PT and INR.**
        - » **Treatment: fresh-frozen plasma, vitamin K.**
    - **Vitamin K deficiency or malabsorption.**
    - **Autoimmune disease.**
    - **Disseminated intravascular coagulation.**

Anticoagulant treatment

- **Anticoagulants are given mainly to prevent and treat thromboembolic disease, also used in atrial fibrillation, after cardiac surgery or organ transplants.**
- **Drugs:**
  - **Warfarin (vitamin K antagonist) for long-term treatment.**
    - **The maximum effect of warfarin takes several day (begin 8–12 h, maximal at 36 h, and persist for 72 hs) , so heparin is given first.**
    - **Prolonging international normalized ratio [INR].**

- Heparin (subcutaneous low-molecular-weight heparin (LMWH) for short-term treatment.
- Others: Dabigatran, rivaroxaban.

- Plasminogen activators such as t-PA (alteplase) or streptokinase also are useful for controlling coagulation.
- Aspirin inhibits platelet aggregation.

Dental Aspects

- **Some patients on anticoagulant drugs have a high tendency to bleed after surgery.**
- **Dental preventive care important to minimize the need for surgical intervention.**
- **Drugs that might worsen bleeding tendency (aspirin and NSAIDs) should be avoided.**
- **Intramuscular injections should be avoided.**
- **Regional block LA injections may also be a hazard.**
- **Intraligamentary or intrapapillary injections are safer and preferred over regional blocks if the INR is above 4.**

- Administration of LA containing vasoconstrictors assists in maintaining a dry surgical field.
- Conscious sedation can be given, If IV care must be taken.
- GA requires expert attention in hospital.
- Anticoagulation should not be stopped, because it may cause thrombotic deaths in dental patients.

- Warfarin does not stopped before primary care and minor dental procedures such as administration of infiltration LA and removable prosthetic appliances. Invasive procedures as the administration of LA via an inferior alveolar nerve block, tooth extractions, periodontal surgery, biopsies as well as the prescribing of certain medications may need modification.

- **Assurance that the patient not also receiving antiplatelet agents (aspirin and/or clopidogrel).**
- **Minimize the risk and extent of postoperative bleeding, the number of teeth to be extracted at any one sitting should be limited.**
- **Warfarin therapy may be reinstituted on the evening of surgery if hemostasis has been assured.**

- **Most cases of post-operative bleeding are easily treated with local measures such as packing with a hemostatic dressing, suturing and pressure.**
- **For elective dental procedures in patient who are taking warfarin for a limited period (6 months for DVT), the procedure should be postponed until the warfarin has been stopped.**

Guidelines of ULMI

- **UK Medicines Information (UKMI) summarizes these guidelines as follows:**
  - **An INR should be measured within 24–72 h of undertaking the procedure.**
  - **The UKMI recommends that no individuals should have a dental procedure in primary care if they have an INR > 4; However the British Committee for Standards in Hematology guidelines recommend that the INR should be no greater than 2 at the time of the procedure.**
  - **The UKMI also recommends that warfarin patients with renal or hepatic disease or those on chemotherapy or cytotoxics should have dental procedures in hospital.**

- **LMWH may have little effect on post-operative bleeding, despite their longer activity (up to 24 h). However, the advice of the hematologist should be sought before surgery.**
- **Heparin has an immediate effect on blood clotting but acts for only 4–6 h and no specific treatment is therefore needed to reverse its effect.**

- **In an emergency, this can be reversed by intravenous protamine sulfate given in a dose of 1 mg per 100 IU heparin, but a medical opinion should be sought first.**
- **Heparin has been stopped, any surgery can safely be carried out after 6–8 h**
- **LMWH is administered 24 h before surgery. Then given on the 2nd day post-operative when hemostasis has been secured.**
- **In practice, aspirin rarely interferes significantly with dental surgical procedures.**

- Which of the following platelet level is considered safe for major surgical procedures?
- A. More than 30 X 109 / L.
- B. More than 45 X 109 / L.
- C. More than 50 X 109 / L.
- D. More than 75 X 109 / L

- Which of the following substances is formed by secondary hemostasis?
- A. Platelet plug.
- B. Fibrin.
- C. Plasmin.
- D. von Willibrand factor

- Preventive dentistry should be started as early as possible in which of the following conditions?
- A. Hemophilia A.
- B. Immune thrombocytopenic purpura.
- C. Hereditary hemorrhagic telangiectasia.
- D. von Willibrand disease.

- **A 70-year-old patient presented by tooth ache. He had thromboembolic disease and was on anticoagulant therapy. Which of the following strategies is correct during his dental management?**
- A. Conscious sedation can be given orally.
- B. Safe to stop his anticoagulant medication at any time.
- C. Intramuscular injection of analgesics is preferred.
- D. Regional nerve block is safe to use

Thank You